



item	example 2
open-loop form	$G(s) = K(2s + 1) / [(s + 1)^2 (\frac{4}{7}s - 1)]$
root-locus gain	$K^* = 3.5K$
imaginary-axis crossings	$\omega = 0, K = 1 \text{ and } \omega = \pm\sqrt{2}, K = 9/7$
design conclusion	the closed loop is only conditionally stable, and the usable range is $1 < K < 9/7$